MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5686

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentode, minature, rf beam power

Outline --- 6-2 (EIA)

Base --- E9-1

Envelope --- T6-1/2

Cathode --- Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	k, g3	g1	k, g3	h	h	g2	a	k, g3	g2

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec2	Ec1	Ehk	Rg1	Ib	Ic1
Unit:	V	Vdc	Vdc	Vdc	v	Meg	mAdc	mAdc
Maximum:	6.9	275	275	0	100	(See note 1)	44	3.3
Minimum:	5.7	---	---	-165	---	---	---	---
TEST CONDITIONS:	6.3	250	250	-12.5	---	---	---	----

Parameter:	TE	Pp	Pg2	Pi	Alt
Unit:	°C	W	W	W	ft
Maximum:	175	8.0	3.3	14.0	(See note 2)
Minimum:	---	---	---	----	
TEST CONDITIONS:	---	---	---	----	----

GENERAL:

Qualification - Required

Reliable tube

(E) denotes changes

FSC 5960

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TABLE 1 Quality Conformance Inspection.

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Qualification Inspection							
1236	Power oscillation (1)	Ec1 = -50 Vdc; Esig/Ib = 40 mAdc; F = 125 MHz t = 20 See note 4	---	---	Po	4.3	---	W
	Quality conformance inspection, part 1							
1231	Emission	Eb = Ec2 = Ec1 = 30 Vdc See note 3	0.4	II	Is	70	---	mAdc
1256	Electrode current (anode)		0.4	II	Ib	21	35	mAdc
1256	Electrode current (1) (screen)		0.4	II	Ic2	---	6.0	mAdc
1266	Total grid current	Rg = 0.02 Meg See note 3	0.4	II	Ic1	0	-2.0	µAdc
1301	Heater current		0.4	II	If	320	380	mA
1336	Heater-cathode leakage		0.4	II	Ihk	---	20	µAdc
1341	Power output (1)	Esig = 8.8 Vac; Rp = 9,000 ohms	0.4	II	Po	2.2	---	W
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1211	Insulation of electrodes		2.5	I	---	---	---	---
1031	Low-frequency vibration	Rp = 2,000 ohms; 10G; F = 40 Hz	---	---	Ep	---	400	mVac
1236	Power oscillation	Esig/Ic1 = 2 mAdc; Ib = 40 mAdc; F = 5 MHz; Rc1 = 25,000 ohms See note 4	---	---	Po	5.25	---	W
1266	Grid emission	Ef = 7.5 V; Ec1 = -45 Vdc; Rg = 0.02 Meg See note 5	---	---	Ic1	0	-2.0	µAdc
1306	Transconductance		---	---	Sm	2,600	4,000	µmhos
1341	Power output (2)	Ef = 5.7 V; Esig = 8.8 Vac; Rp = 9,000 ohms	2.5	I	ΔPo Ef	---	15	%
1256	Electrode current (2) (screen)	Power oscillation (2)	---	---	Ic2	---	13	mAdc

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TABLE 1 Quality Conformance Inspection (Continued).

METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS Min	LIMITS Max	UNIT
	Quality conformance inspection, part 2 -Continued							
1331	Direct- interelectrode capacitance	Shield No. 315 Shield No. 315 Shield No. 315	6.5	Code E	C glp C in C out	--- 5.0 7.0	0.08 8.0 10.0	pF pF pF
1041	Shock	450G; Ehk = 100 Vdc; Ec1 = -6.5 Vdc; Rk = 150 ohms See note 7	---	---	---	---	---	---
1031	Vibration fatigue	2.5 G; fixed frequency; F = 25 min, 60 max	6.5	Note 6	---	---	---	---
----	Post-shock and vibration-fatigue test end points:							
1031	Low-frequency vibration		---	---	Ep	---	600	mVac
1336	Heater cathode leakage		---	---	Ihk	---	50	µAdc
1306	Transconductance		---	---	Sm	2,200	---	µmhos
1266	Total grid current		---	---	Ic1	---	-4.0	µAdc
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
	Quality conformance inspection, part 3							
1506	Heater-cycling life	Ef = 7.5 Vac; Ehk = 135 Vdc; Eb = Ec1 = Ec2 = 0			---	---	---	---
----	Heater-cycling life test end point:							
1336	Heater-cathode leakage				Ihk	---	40	µAdc
1516	Stability life	Ec1 = -6.5 Vdc; Ehk = 180 Vdc; Rg1 = 0.01 Meg; Rk = 150 ohms; TA = room			---	---	---	---
----	Stability-life test end point:							
1341	Power output (1)	Change in power output (1) of individual tubes			ΔPo t	---	15	%

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TABLE I Quality Conformance Inspection (Continued).

Ⓔ METHOD	REQUIREMENT OR TEST	CONDITIONS	Sym	LIMITS Min	LIMITS Max	Unit
	Quality conformance inspection, part 3 -Continued					
1501	Intermittent life	Stability life-test conditions; TE = 175°C (min) See note 8	---	---	---	---
	Intermittent life-test end points: (500 hours)					
----	Inoperatives		---	---	---	---
1266	Total grid current		Icl	---	-2.0	µAdc
1236	Power oscillation (2)		Po	4.75	---	W
1341	Power output (2)		ΔPo Ef	---	15	%
1341	Power output (1)	Change in power output (1) of individual tubes	ΔPo t	---	15	%
1336	Heater-cathode leakage		Ihk	---	30	µAdc
1211	Insulation of electrodes		R	60	---	Meg
1341	Power output (1)	Average change	Avg ΔPo t	---	10	%
----	Total defectives		---	---	---	---
----	Intermittent life-test end points: (1,000 hours)					
----	Inoperatives		---	---	---	---
1266	Total grid current		Icl	---	-2.0	µAdc
1236	Power oscillation (2)		Po	4.25	---	W
1341	Power output (2)		ΔPo Ef	---	20	%
1341	Power output (1)	Change in power output (1) of individual	ΔPo t	---	20	%
1336	Heater-cathode leakage		Ihk	---	40	µAdc
1211	Insulation of electrodes		R	50	---	Meg
----	Total defectives		---	---	---	---
1511	Cathode-interface life	See note 9	---	---	---	---
----	Cathode-interface life-test end points:	Ef = 5.7 ±0.05 Vdc; Eb = 80 Vdc; Ec/Ib = 5.0 mAdc See note 10	Ri	---	50	ohms

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NOTES:

1. The following Rg1 values shall apply:

	With fixed bias	With cathode bias
Class A1, amplifier	0.10	0.50
Class C, rf amplifier	0.05	0.05

2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.

3. This test is to be performed at the conclusion of the holding period.

4. In a class C amplifier circuit, Po shall be useful power output.

5. Prior to this test, tubes shall be preheated a minimum of 5 minutes with each section operating at conditions specified below. The 3-minute test is not permitted. Test at specified conditions within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid-emission test.

Ef V	Ec1 Vdc	Ec2 Vdc	Eb Vdc	Rk ohms	Rg1 Meg
7.5	-4.5	250	250	150	0.01

6. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.

7. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is employed.

8. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a tube having bogey Ib (+5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.

9. The life-test sample shall consist of 10 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, due to failure of more than 1 tube, a second sample of 20 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples. The total tube failures from the combined first and second samples shall not exceed two. A life-test defect is defined as a failure to meet the life-test end-point limits specified in the tube specification sheet.

10. Preheat approximately 5 minutes prior to testing, using Ef = 5.7 V, other electrodes disconnected. No other test shall be made from the start of the interface life test until after the measurement of the end-point characteristic following completion of the indicated minimum number of life-test hours.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Preparing activity:
Air Force - 85

Review activities:
Army - ER
Air Force - 99

(Project No. 5960-3308)

User activities:
Army - ME, WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

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